Identifiability of species interactions in a living material: Bayesian calibration of an extended Hamilton principle multiphase model

Keisuke Nishioka[a,b,c,d,*], Felix Klempt[a], Hendrik Geisler[a], Rumjhum Mukherjee[b,c], Nils Heine[b,c], Katharina Doll-Nikutta[b,c], Meike Stiesch[b,c], Szymon P. Szafrański[b,c], Meisam Soleimani[a], Philipp Junker[a]

[a] *Institute of Continuum Mechanics, Leibniz Universität Hannover, An der Universität 1, 30823 Garbsen, Germany*

[b] *Department of Prosthetic Dentistry and Biomedical Materials Science, Hannover Medical School, Hannover, Germany*

[c] *Lower Saxony Centre for Biomedical Engineering, Implant Research and Development (NIFE), Hannover, Germany*

[d] *School of Science for Open and Environmental Systems, Graduate School of Science and Technology, Keio University, Yokohama, Japan*

[*] Corresponding author

Email address: keisuke.nishioka@stud.uni-hannover.de, keisuke58@keio.jp (Keisuke Nishioka)

Abstract

Biofilms are living materials whose composition, and with it their mechanical response, depends on species interactions that cannot be measured directly. We calibrate a thermodynamically consistent multiphase model of a five-species oral biofilm, derived from the extended Hamilton principle, against in-vitro time courses of a commensal and a dysbiotic community, each under static and flow-reactor cultivation. All 15 entries of the symmetric interaction matrix are inferred by transitional Markov chain Monte Carlo, with the stiff forward model compiled in JAX and evaluated for all particles in parallel on a GPU (about 150 times faster than on a CPU). Data enter in stages, from species composition to community viability, and each stage is checked for seed agreement, mixing and posterior mass at the prior bounds. Root-mean-square errors of the species fractions are 0.02–0.08 in three conditions[prov.]; in the commensal static condition (0.17–0.18) composition and measured viability cannot both be matched. The data constrain only part of the interaction matrix: four and six entries in the two commensal conditions, all among the early colonisers and with the same signs where they overlap, and eight and nine in the dysbiotic ones[prov.]. The coefficient between *F. nucleatum* and *P. gingivalis* is constrained in both dysbiotic conditions but not sign-determined[prov.]. The pH predicted from the calibrated composition follows the measured pH in the dysbiotic flow-reactor community (offset about 0.13 in the commensal one), and removing *F. nucleatum* in silico abolishes the late increase of *P. gingivalis*, a prediction that can be tested directly[prov.].

Keywords: Inverse characterization, Bayesian calibration, Uncertainty quantification, Extended Hamilton principle, Oral biofilm, GPU

1. Introduction

Biofilms are soft, living materials: microbial cells embedded in a self-produced matrix whose composition evolves in time and determines their mechanical response [1]. On dental implants, a shift in the species composition of the biofilm from a commensal towards a dysbiotic community is associated with peri-implantitis, and in-vitro models of peri-implant biofilms reproduce such shifts [2]; in the ecological view of oral disease, the shift of the

community, rather than a single pathogen, drives disease [3]. Early biofilms on implant abutments in vivo already contain disease-associated species [4]. Which interactions between the species drive such a shift is difficult to measure: co-culture experiments report how the composition changes over time, not the interactions themselves.

Continuum mechanics offers a route from composition to interaction. Based on the extended Hamilton principle [5], Klempt et al. [6] derived a multiphase model in which species volume fractions and viabilities evolve under a free energy that contains an interaction matrix $\mathbf{A}$, taken symmetric so that the interactions derive from this potential; thermodynamic consistency follows from the variational structure. Calibrating $\mathbf{A}$ against data is an inverse characterization problem: the physics fixes the structure of the model, and the data determine its coefficients. Data-driven methods for such problems are most reliable when they are guided by physics and by explicit checks of what the data can determine. Once calibrated, the model can be used as an in-silico test bed, for example to ask how the community would develop if one species were removed. Bayesian updating has been used to calibrate constitutive parameters of biofilm models with surrogates [7], but the interaction matrix poses two specific difficulties. First, five time points of five species fractions that sum to one give 20 independent composition values, plus five values of the community viability, against 15 unknowns, so identifiability cannot be assumed. Second, the coupled volume-fraction and viability dynamics produce posteriors that are multimodal and bounded by the prior box, for which the convergence of standard samplers is hard to verify.

Here we infer all 15 entries of $\mathbf{A}$ from the time courses of Heine et al. [2] for five oral taxa—*Streptococcus oralis* (So), *Actinomyces naeslundii* (An), *Veillonella* spp. (Vei), *Fusobacterium nucleatum* (Fn) and *Porphyromonas gingivalis* (Pg), numbered 1 to 5 in this order—in four conditions: a commensal or a dysbiotic community, each under static cultivation in well plates or dynamic cultivation in a flow reactor (HOBIC, Hannoverian Oral multispecies Biofilm Implant flow Chamber), abbreviated CS, CH, DS and DH (commensal/dysbiotic, static/HOBIC). We combine three elements: first, transitional Markov chain Monte Carlo (TMCMC) [8, 9] with the forward model compiled in JAX [10] and evaluated for all particles in parallel on a GPU; second, a staged introduction of the data, from species composition to community viability; and third, explicit acceptance criteria for every stage, including a rule for widening prior bounds at which posterior mass accumu-

lates, and an analysis of which interactions the data determine. The data determine only a subset of $\mathbf{A}$, which we identify entry by entry; where the commensal conditions overlap, the identified entries agree in sign across cultivation modes, whereas the Vei–Pg coefficient of the dysbiotic community changes sign between static and dynamic cultivation[**prov.**].

2. Model

At a material point, species $i = 1, \dots, 5$ (So, An, Vei, Fn, Pg) occupies the volume fraction $\phi_i \in (0, 1)$, of which the fraction $\psi_i \in (0, 1)$ is alive; the living volume fraction is $\bar{\phi}_i = \phi_i \psi_i$. A void fraction ϕ_0 closes the volume, $\sum_{l=0}^{5} \phi_l = 1$. As in Klempt et al. [6], the evolution equations result from the stationarity of the rate functional (derivation in Supplementary Section S1)

$$\dot{\Psi} + \Delta + C \to \mathrm{stat}_{(\dot{\boldsymbol{\phi}}, \dot{\psi}, \gamma)}, \tag{1}$$

the local form of the extended Hamilton principle [5, 11, 12, 13], with the free energy, dissipation and constraint

$$\Psi = -\tfrac{1}{2}\, c^* \, \bar{\boldsymbol{\phi}}^{\mathsf{T}} \mathbf{A} \, \bar{\boldsymbol{\phi}}, \qquad \Delta = \tfrac{1}{2}\, \dot{\bar{\boldsymbol{\phi}}}^{\mathsf{T}} \boldsymbol{\eta} \, \dot{\bar{\boldsymbol{\phi}}} + \tfrac{1}{2}\, \dot{\boldsymbol{\phi}}^{\mathsf{T}} \boldsymbol{\eta} \, \dot{\boldsymbol{\phi}}, \qquad C = \gamma \Big(\sum_{l=0}^{5} \phi_l - 1 \Big). \tag{2}$$

Here c^* is the nutrient concentration, $\boldsymbol{\eta} = \mathrm{diag}(\eta_i)$ with $\eta_i = 1$, and γ is a Lagrange multiplier; the antibiotic term of the original model vanishes because the experiments contain no antibiotic. For $n = 5$ species, stationarity gives $2n + 2 = 12$ coupled non-linear equations, for $i = 1, \dots, 5$,

$$0 = -c^* \psi_i (\mathbf{A} \bar{\boldsymbol{\phi}})_i + \eta_i \Big(\dot{\phi}_i \psi_i^2 + \bar{\phi}_i \dot{\psi}_i + \dot{\phi}_i \Big) + \gamma, \tag{3}$$

$$0 = -c^* \phi_i (\mathbf{A} \bar{\boldsymbol{\phi}})_i + \eta_i \Big(\dot{\psi}_i \phi_i^2 + \bar{\phi}_i \dot{\phi}_i \Big) + \gamma, \tag{4}$$

together with $0 = \gamma + \dot{\phi}_0$ and the volume constraint. They are integrated by an implicit Euler scheme with Newton iterations (2500 steps covering the 21-day experiment), with a logarithmic barrier keeping ϕ_i, ψ_i inside $(0, 1)$. Only the symmetric part of $\mathbf{A}$ enters the quadratic free energy, so $\mathbf{A}$ is taken symmetric, with 15 independent entries, which form the parameter vector $\boldsymbol{\theta} \in \mathbb{R}^{15}$. The coefficient a_{ij} is an effective, undirected interaction of the model; it lumps mechanisms that the model does not resolve, such as metabolite exchange or co-aggregation. Only the product $c^* a_{ij}$ enters, so we fix $c^* = 25$ to keep $a_{ij} = \mathcal{O}(1)$. Removing a species is represented by holding its volume fraction at zero.

3. Bayesian inference

Likelihood.. The composition data are the species fractions of viable cells at days 3, 6, 10, 15 and 21 (day 1 is the initial condition), measured by quantitative real-time PCR after propidium monoazide (PMA) treatment [2], with the total-count fractions without PMA measured on the same samples (their Figs. 3A and S1). We use the measured replicate values provided by the authors (three to five biological replicates per time point, each in technical triplicate) and the medians over them; values read off the published box plots deviate from these by up to 0.42 in a species fraction in the dysbiotic conditions, while the live fraction and the biofilm volume are read off to within 0.015 and a few percent (Supplementary Section S10). Model time is the time since inoculation, $t = s\,d$ with $s = 0.2375/21$; the day-1 composition, the earliest measurement, serves as the initial state at $t = 0$. In the flow reactor the inoculated medium is replaced by sterile 1:2-diluted medium after 24 h [2], so the growth phase there can also be taken to start at day 1. Starting the model at day 1 ($t = s\,(d-1)$) changes the composition error of the maximum a posteriori estimate (MAP) by at most 0.008 in any condition without re-inference**[prov.]**. **[2c: this paragraph and Eq. (lik) are replaced by the two-channel likelihood once the raw-data runs are in.]** In these data the viable and the total-count compositions differ (root-mean-square error, RMSE, between their medians 0.101, 0.087, 0.119 and 0.087 in CS, CH, DS and DH; Supplementary Section S11), so we assume species-independent viability in the composition channel (the data show it to be violated mainly for the rare Fn and Pg in the commensal community; Supplementary Section S12) and compare the data with the model volume fractions normalised over the five species, $\hat{y}_i = \phi_i / \sum_j \phi_j$, through a weighted Gaussian log-likelihood

$$\ln p(\mathbf{y}_{\mathrm{obs}} \mid \boldsymbol{\theta}) = -\frac{1}{2} \sum_{k} \sum_{i=1}^{5} w_{ik} \, \frac{\big(y_{\mathrm{obs},i}(t_k) - \hat{y}_i(t_k;\boldsymbol{\theta})\big)^2}{\sigma_i^2} + \mathrm{const}, \tag{5}$$

with σ_i the standard deviation (SD) of species i at each time point estimated from the spread of the measured replicates (interquartile range/1.35 over all measurements, at least 0.05). The weights up-weight Pg ($\lambda_{\mathrm{Pg}} = 5$), whose late increase is the main dynamic feature of the dysbiotic data, and the last two time points ($\lambda_{\mathrm{late}} = 3$), and down-weight species with a mean fraction below 5% ($\lambda_{\mathrm{rare}} = 0.1$; Supplementary Table S2); with $\lambda_{\mathrm{Pg}} = \lambda_{\mathrm{late}} = 1$

(λ_{rare} unchanged) the median of a_{45} in DH is unchanged but its interval widens (Supplementary Section S9). In stages (iii) and (iv) a second Gaussian channel compares the predicted community live fraction $\sum_i \bar{\phi}_i / \sum_i \phi_i$ with the measured fraction of membrane-intact cells from SYTO9/propidium iodide (LIVE/DEAD BacLight) staining, which is image based and not resolved by species (mean over 15–45 confocal images per time point, provided by the authors); it is weighted by $\lambda_\psi = 1.5$ in the commensal conditions, 2.0 in DS and 3.0 in DH (Supplementary Table S2). The measured pH under dynamic cultivation is not used in the inference and serves as a check (Section 4).

Prior and stages.. The prior is uniform on a box $[l_k, u_k]$ for each entry. The stage-(i) boxes were set from earlier calibrations of a previous version of the model on the same data, and some of them exclude negative values or allow only small ones (CS: $a_{11} \in [0, 2]$; DH: $a_{11}, a_{22}, a_{44} \in [0, 5]$, $a_{12} \in [0.5, 3]$, $a_{23} \in [0, 3]$, $a_{45} \in [-0.5, 6]$), so the signs of these entries are partly set by the prior in stages (i)–(ii); all of these bounds were widened in stage (iii) (Supplementary Table S4). Inference proceeds in four stages on the same data (Fig. 1): (i) composition likelihood on the box; (ii) as (i), on the box narrowed to the stage-(i) posterior mean $\pm 4\,\mathrm{SD}$; (iii) composition and viability, back on the full box, starting from the stage-(ii) particles; (iv) as (iii), on the box narrowed to the stage-(iii) posterior mean $\pm 4\,\mathrm{SD}$. A $\pm 4\,\mathrm{SD}$ box removes less than 10^{-4} of a Gaussian marginal, so narrowing shortens the tempering without truncating the posterior. If more than 20% of the posterior mass of an entry lies in the outer 5% of the box on one side, or this fraction grows by more than 0.10 from the previous stage, the bound is moved outwards and the stage is repeated; the boxes finally used are listed in Supplementary Table S4.

TMCMC on the GPU.. TMCMC [8, 14] passes a particle population through the tempered posteriors $p_m(\boldsymbol{\theta}) \propto p(\mathbf{y}_{\mathrm{obs}} \mid \boldsymbol{\theta})^{\beta_m} p(\boldsymbol{\theta})$, $0 = \beta_0 < \cdots < \beta_K = 1$ (K tempering steps), choosing each increment so that the coefficient of variation of the importance weights stays at a target value [9]. After resampling, every particle performs n_{mut} Metropolis–Hastings steps with random-walk proposals scaled by the weighted particle covariance, alternating with differential-evolution proposals [15] that follow strongly correlated directions; proposals outside the box are rejected without solving the model. Each likelihood evaluation requires one solution of Eqs. (3)–(4), and a run requires 10^5 to 3×10^7 of them. We implement the implicit solver, including Jacobian

assembly and linear solve, in JAX, compile it with the XLA (Accelerated Linear Algebra) compiler into a single kernel and evaluate all particles at once with `vmap`. On one GPU (NVIDIA RTX 4090) this is 170 times faster per tempering step than a 12-core CPU at matched settings (147 times end to end; Supplementary Table S3), which makes it affordable to repeat every stage with three seeds.

Acceptance criteria.. Every stage is run with three independent seeds and accepted only if (a) the stored log-likelihood of every particle agrees with a recomputation; (b) the tempering reaches $\beta = 1$; (c) each particle is moved at least twice per tempering step and at least five times per free parameter over the run; (d) the maximum log-likelihood, refined by a local optimisation (L-BFGS-B) from the best particles of each seed, differs by at most 1 across seeds, i.e. all seeds reach the same likelihood peak; and (e) every posterior median differs across seeds by at most half a posterior SD; an entry whose pooled posterior SD is at least 0.8 times that of a uniform distribution on the prior box of the stage is reported as not identified and excluded from this criterion, and for an entry with several posterior modes the criterion is applied within each mode, with the mode weights reported together with their spread across seeds. We call all other entries identified: the data constrain them beyond their prior box. Criterion (c) counts accepted moves rather than the acceptance rate: with optimally scaled random-walk proposals [16], $5d$ accepted moves displace each coordinate by about five posterior standard deviations, independently of the dimension d. Stages (ii)–(iv) that fail are repeated with more particles or mutation steps and not interpreted. Stage (i) is exploratory; it sets the stage-(ii) box and gives the evidence estimates of TMCMC, which are used only to compare models on the same data and box and are reported with the caveat that not all stage-(i) groups meet the criteria (Section 4).

4. Results and discussion

4.1. Fit and convergence

Stages (ii)–(iv) of CS and CH and stages (ii)–(iii) of DS meet criteria (a)–(e) with three seeds each; the DH stage-(iii) group meets (a)–(d), but its seed spread of the medians reaches 0.64 posterior SD (criterion (e): 0.5)**[DH: larger run in progress; DS, DH stage (iv) pending; their numbers below are from stage (iii)]**. Of the exploratory stage-(i) groups, CS and

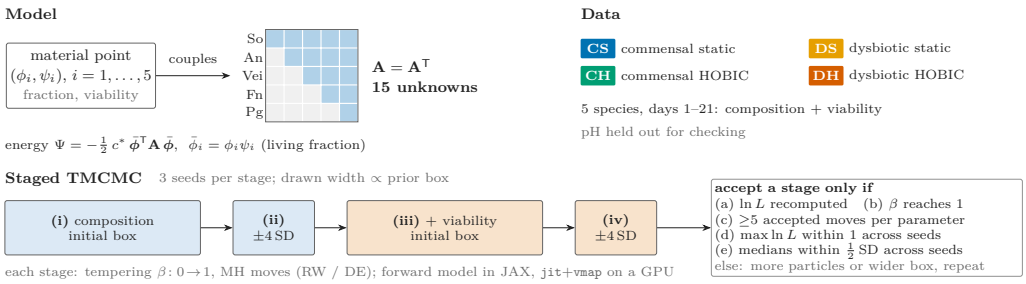


Figure 1: Model and inference. A material point carries the volume fractions ϕ_i and viabilities ψ_i of five taxa, coupled through the symmetric interaction matrix $\mathbf{A}$ (15 unknowns). TMCMC with a GPU-compiled forward model infers $\mathbf{A}$ in four stages, from species composition to community viability; every stage is accepted only after the checks (a)–(e).

CH do not meet (c) (4.4 and 3.8 moves per free parameter) and DS does not meet (e) (Supplementary Table S5). At the final stage the composition is reproduced with an RMSE of 0.076–0.078 in CH, 0.172–0.179 in CS, 0.020–0.028 in DS and 0.081–0.083 in DH[prov.] (Fig. 2). Accepting a stage often required more than one attempt: posterior mass accumulated at a bound of the initial box in every condition and was released by widening that bound (Supplementary Table S4), and the dysbiotic conditions needed [18000 (DS) and 14000 (DH)] particles in stage (iii) (Supplementary Table S5). Without the acceptance criteria, several of these runs would have been read as converged.

4.2. What the data determine

Figure 3 and Supplementary Table S7 classify every entry of $\mathbf{A}$ at the final stage, and Supplementary Fig. S2 shows the marginal posteriors; the number of directions in which the likelihood itself is narrow at the MAP is 4, 4, 8 and 5, against 4, 6, 8 and 9 identified entries (Supplementary Section S6); the counts depend on the threshold 0.8, the pattern described below does not (Supplementary Table S6). In the commensal conditions only a few entries are identified: four in CS (a_{11}, a_{22}, a_{33}, a_{13}) and six in CH (a_{11}, a_{12}, a_{22}, a_{33}, a_{13}, a_{23}), all among the early colonisers So, An and Vei. Where both conditions identify an entry, the sign agrees: the self-interaction of So is negative (a_{11}, 90% intervals $[-4.9, -1.2]$ in CS and $[-2.5, -0.5]$ in CH) and the So–Vei coefficient is positive (a_{13}, $[+2.8, +9.2]$ and $[+0.01, +1.4]$), in line with the lactate cross-feeding between streptococci and *Veillonella* [17],

although a_{13} is an effective coefficient and not a measurement of that mechanism. Two further entries of CH, a_{14} and a_{15}, are determined in sign only (90% intervals $[-4.8, -0.2]$ and $[-4.8, -0.3]$), and every other entry involving Fn or Pg spreads over its whole box. This is expected: Fn and Pg stay below 1% of the commensal communities from day 3 on, so the composition data carry little information on their interactions. In the dysbiotic conditions, where Fn and Pg reach 10–27% of the community, more entries are identified: eight in DS and nine in DH, now including interactions of the late colonisers**[prov.]**. The Vei–Pg coefficient a_{35} is identified in both but with opposite signs, positive under static cultivation (DS, $[+0.3, +2.4]$) and negative in the flow reactor (DH, $[-13.4, -1.7]$), and the An–Fn coefficient a_{24} is positive in DH ($[+2.7, +7.1]$). A sign change between cultivation modes is a statement about the effective model under different nutrient and flow conditions, not about the pairwise mechanism itself. Comparisons between the commensal and the dysbiotic community are confounded by design: the two communities differ not only in their ecological state but also in the *Veillonella* species (*V. dispar* vs. *V. parvula*) and the Pg strain (DSM 20709 vs. W83) [2], so differences between their matrices combine state, strain physiology and cultivation; the DS–DH contrast above compares the same strains and is free of this confound, although the two conditions also differ in the viability weight ($\lambda_\psi = 2.0$ vs. 3.0) and in their prior boxes.

The classification is a result in its own right: it states which conclusions about the interaction network the data can support, and it is obtained from the same runs at no additional cost.

4.3. The Fn–Pg coefficient

The coefficient a_{45} links the bridging organism Fn to the pathogen Pg. Comparing the full model with a model in which $a_{45} = 0$, both inferred at stage (i), gives $\ln B = +0.10$ (CS), $+0.05$ (CH), -2.67 (DS) and $+1.94$ (DH) (Supplementary Table S8); because not all stage-(i) groups meet the acceptance criteria, these values are indicative. Because the uniform prior penalises the full model by the log-width of the a_{45} box, $\ln B$ is read only within a condition: in DS, where the stage-(i) box is $[-15, 20]$, the penalty alone is $\ln 35 \approx 3.6$. In DH, the data favour a non-zero coefficient. With the composition alone (stage (i)), a_{45} in DH is positive in all seeds (90% interval about $[+0.5, +5.5]$) within the stage-(i) box, which allows only small negative values ($[-0.5, 6]$); with $\lambda_{\mathrm{Pg}} = \lambda_{\mathrm{late}} = 1$ its median is unchanged but the lower 5% bound drops to -0.06 to $+0.17$ across seeds (Supplementary

Section S9). Once the viability enters, on a box widened to $[-5, 20]$ in DH, the median of a_{45} stays positive in both dysbiotic conditions (+2.8 in DS, +1.2 in DH) but the 90% intervals include zero ($[-1.3, +15.1]$ and $[-1.9, +7.5]$; Supplementary Fig. S2)[prov.]: a_{45} is constrained but not sign-determined. Within the stage-(i) box, the composition data thus favour a positive Fn–Pg coefficient in DH, and the viability data neither confirm nor contradict it.

4.4. Checks against data not used in the inference

The pH measured under dynamic cultivation was not used in the inference (Fig. 4b,c). A linear regression of the measured pH on the measured species fractions (intercept and four fractions, Vei as reference; $R^2 = 0.93$ over 90 time points in CH and DH) was fitted once on the measured data and then applied, unchanged, to the posterior predictive composition, so the check tests the predicted composition, not a model of pH. In DH the pH predicted in this way follows the measured rise from about 6.3 to 7.0 ($R^2 = 0.63$, RMSE 0.12)[prov.]; in CH it is offset upwards by about 0.13, because the model keeps Fn and Pg near 4–5% where the data show below 1%, likely a consequence of down-weighting rare species in the likelihood, and these two taxa carry the largest regression coefficients. The calibrated model can also be used in silico: removing Fn from the dysbiotic flow-reactor community abolishes the late increase of Pg (Fig. 4a). The day-21/day-15 ratio of Pg falls from 2.9 (90% interval $[2.3, 3.6]$; measured 2.74) to 1.1 ($[0.3, 1.8]$), and only 2–7% of the posterior samples keep a ratio of 1.5 or more[prov.]. Setting only $a_{45} = 0$ removes the increase in most but not all samples (17–52% keep it, depending on the seed): the effect of a_{45} alone varies between seeds, whereas the effect of removing Fn is consistent across seeds. The prediction therefore rests mainly, though not only, on the Fn–Pg coefficient; it can be tested directly by a co-culture without Fn. Under static cultivation the model shows no comparable late increase (ratio 1.1), so the knockout is not informative there.

4.5. Limitations

The likelihood treats the five species fractions as independent although they sum to one, so the posterior intervals may be somewhat too narrow. The fit is poorest in CS, almost entirely in So and Vei (per-species errors 0.31 and 0.24, all others below 0.05), where the So fraction rises from 0.18 to 0.70 between day 1 and day 3. The model can follow this rise: a deterministic

optimisation of the composition error alone, started from the MAP with every entry in $[-15, 20]$, reaches an error of 0.075, but only with a live fraction near 0.97, against the measured 0.65–0.77 (Supplementary Section S13). The composition and the live/dead staining cannot both be matched in CS, and the posterior settles on a compromise; we report it rather than re-weight the channels for this condition. The compromise is sharpest at the MAP: the bulk of the CS posterior lies closer to the composition data (pointwise posterior median, error 0.125), whereas the MAP, nearly identical in all three seeds (composition error 0.172–0.179), sits where the live fraction is matched best; in the other conditions the two differ by less than 0.02 (Supplementary Section S7). The community live fraction does not determine the species-specific viabilities ψ_i (in the model they span 0.03–0.96), so the living fractions of individual species are not predictions of the calibrated model; only the community live fraction is (Supplementary Fig. S4a). Viable and total-count compositions used as separate channels would constrain ψ_i species by species (Supplementary Section S12). The composition data are viable-cell fractions; the model fractions are about as close to the total-count data (Supplementary Section S11). The interaction coefficients are effective quantities of this model and data set; they lump mechanisms that the model does not resolve, such as metabolite exchange and pH, and are not direct measurements of pairwise mechanisms. The calibration covers a 0D material point; the spatial structure of the biofilm and its mechanics are not part of the inference. The calibrated dynamics are, however, the input that continuum growth models of biofilms require [1]: the total volume fraction of the point model can drive the growth variable of a finite-element model, so that differences in growth-induced stress between conditions follow from the calibrated interactions rather than from assumed species-specific growth weights. Because the likelihood uses species fractions only, the absolute amount of biomass is a prediction of the model rather than a calibrated quantity, and its posterior spread has to be propagated into such a coupling. At a material point constrained in all directions, growth driven by the living volume separates the conditions, whereas growth driven by the total volume, which the data do not constrain, does not (Supplementary Section S14).

5. Conclusions

The interaction matrix of a variational multiphase biofilm model was calibrated against time courses of a five-species oral community in four con-

ditions, with a GPU-compiled forward model, a staged likelihood and explicit acceptance criteria for every stage. Three points carry over to other inverse characterizations of living materials. First, convergence has to be checked, not assumed: posterior mass at the prior bounds was found in every condition and disagreement between seeds in the dysbiotic ones, and both were addressed only by widening bounds and adding particles. Second, identifiability is a result: with 20 independent composition values and five viability values for 15 unknowns the data determine only some interactions, and stating which ones limits the biological conclusions that can be drawn from the model. Third, a misfit can be informative: in the commensal static condition the composition and the viability data cannot be matched at once, which may point to a process the model lacks, or to a mismatch between the live/dead stain and the model viability, rather than to a failed calibration. In the dysbiotic conditions more interactions are identified, and the Fn–Pg coefficient, which the composition data favour as positive in DH within the stage-(i) box, is constrained but not sign-determined by the full data set**[prov.]**.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Claude (Anthropic) in order to improve the language and readability of the manuscript and to assist with programming the analysis and plotting scripts. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Data and code availability

Code, run configurations, convergence records and posterior samples are archived at Zenodo (doi:10.5281/zenodo.23320423). The experimental data are those of Heine et al. [2] and are available from those authors; they are not redistributed with the archive.**[update after N. Heine's reply on including the replicate tables]**

Funding

This work was funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) under the Collaborative Research Center SFB/TRR-298-SIIRI – Project ID 426335750.

Funded by the European Union (ERC, Gen-TSM, project number 101124463). Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the European Research Council Executive Agency. Neither the European Union nor the granting authority can be held responsible for them.

Acknowledgements

Computational resources of the Multiphysics Materials Computation Research Group (Assoc. Prof. M. Muramatsu), Department of Mechanical Engineering, Keio University, are gratefully acknowledged.

CRediT authorship contribution statement

[draft, to be confirmed by all co-authors]

Keisuke Nishioka: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft.

Felix Klempt: Methodology, Software, Validation, Writing – review & editing.

Hendrik Geisler: Validation, Writing – review & editing.

Rumjhum Mukherjee: Validation, Writing – review & editing.

Nils Heine: Resources, Data curation, Writing – review & editing.

Katharina Doll-Nikutta: Resources, Writing – review & editing.

Meike Stiesch: Resources, Funding acquisition, Writing – review & editing.

Szymon P. Szafrański: Validation, Writing – review & editing.

Meisam Soleimani: Conceptualization, Methodology, Supervision, Project administration, Writing – review & editing.

Philipp Junker: Conceptualization, Supervision, Funding acquisition, Writing – review & editing.

References

[1] F. Klempt, M. Soleimani, P. Wriggers, P. Junker, A Hamilton principle-based model for diffusion-driven biofilm growth, Biomechanics and Modeling in Mechanobiology 23 (2024) 2091–2113. doi:10.1007/s10237-024-01883-x.

[2] N. Heine, K. Bittroff, S. P. Szafrański, M. Duitscher, W. Behrens, C. Vollmer, C. Mikolai, N. Kommerein, N. Debener, K. Frings, A. Heisterkamp, T. Scheper, M. L. Torres-Mapa, J. Bahnemann, M. Stiesch, K. Doll-Nikutta, Influence of species composition and cultivation condition on peri-implant biofilm dysbiosis in vitro, Frontiers in Oral Health 6 (2025) 1649419. doi:10.3389/froh.2025.1649419.

[3] P. D. Marsh, Are dental diseases examples of ecological catastrophes?, Microbiology 149 (2003) 279–294. doi:10.1099/mic.0.26082-0.

[4] S. Dieckow, S. P. Szafrański, J. Grischke, T. Qu, K. Doll-Nikutta, M. Steglich, I. Yang, S. Häussler, M. Stiesch, Structure and composition of early biofilms formed on dental implants are complex, diverse, subject-specific and dynamic, npj Biofilms and Microbiomes 10 (2024) 155. doi:10.1038/s41522-024-00624-3.

[5] P. Junker, D. Balzani, An extended Hamilton principle as unifying theory for coupled problems and dissipative microstructure evolution, Continuum Mechanics and Thermodynamics 33 (2021) 1931–1956. doi:10.1007/s00161-021-01017-z.

[6] F. Klempt, H. Geisler, M. Soleimani, P. Junker, A continuum multispecies bacterial growth model with a novel interaction scheme, Archive of Applied Mechanics 96 (2026) 164. doi:10.1007/s00419-026-03160-y.

[7] L. Fritsch, H. Geisler, J. Grashorn, F. Klempt, M. Soleimani, M. Broggi, P. Junker, M. Beer, Bayesian Updating of constitutive parameters under hybrid uncertainties with a novel surrogate model applied to biofilms, Computational Mechanics 78 (2026) 761–782. doi:10.1007/s00466-026-02777-8.

[8] J. Ching, Y.-C. Chen, Transitional Markov chain Monte Carlo method for Bayesian model updating, model class selection, and model averaging, Journal of Engineering Mechanics 133 (2007) 816–832.

[9] W. Betz, I. Papaioannou, D. Straub, Transitional Markov chain Monte Carlo: Observations and improvements, Journal of Engineering Mechanics 142 (2016) 04016016.

[10] J. Bradbury, R. Frostig, P. Hawkins, M. J. Johnson, C. Leary, D. Maclaurin, G. Necula, A. Paszke, J. VanderPlas, S. Wanderman-Milne, Q. Zhang, JAX: Composable transformations of Python+NumPy programs, 2018. URL: https://github.com/google/jax.

[11] P. Junker, T. Bode, K. Hackl, On a holistic variational formulation for material modeling including dissipative evolution, Journal of the Mechanics and Physics of Solids 200 (2025) 106133. doi:10.1016/j.jmps.2025.106133.

[12] P. Junker, J. Makowski, K. Hackl, The principle of the minimum of the dissipation potential for non-isothermal processes, Continuum Mechanics and Thermodynamics 26 (2014) 259–268. doi:10.1007/s00161-013-0299-4.

[13] K. Hackl, Generalized standard media and variational principles in classical and finite elastoplasticity, Journal of the Mechanics and Physics of Solids 45 (1997) 667–688. doi:10.1016/S0022-5096(96)00110-X.

[14] P. Del Moral, A. Doucet, A. Jasra, Sequential Monte Carlo samplers, Journal of the Royal Statistical Society: Series B 68 (2006) 411–436.

[15] C. J. F. ter Braak, A Markov chain Monte Carlo version of the genetic algorithm Differential Evolution: easy Bayesian computing for real parameter spaces, Statistics and Computing 16 (2006) 239–249. doi:10.1007/s11222-006-8769-1.

[16] G. O. Roberts, J. S. Rosenthal, Optimal scaling for various Metropolis–Hastings algorithms, Statistical Science 16 (2001) 351–367. doi:10.1214/ss/1015346320.

[17] P. E. Kolenbrander, R. J. Palmer Jr., S. Periasamy, N. S. Jakubovics, Oral multispecies biofilm development and the key role of cell–cell distance, Nature Reviews Microbiology 8 (2010) 471–480.

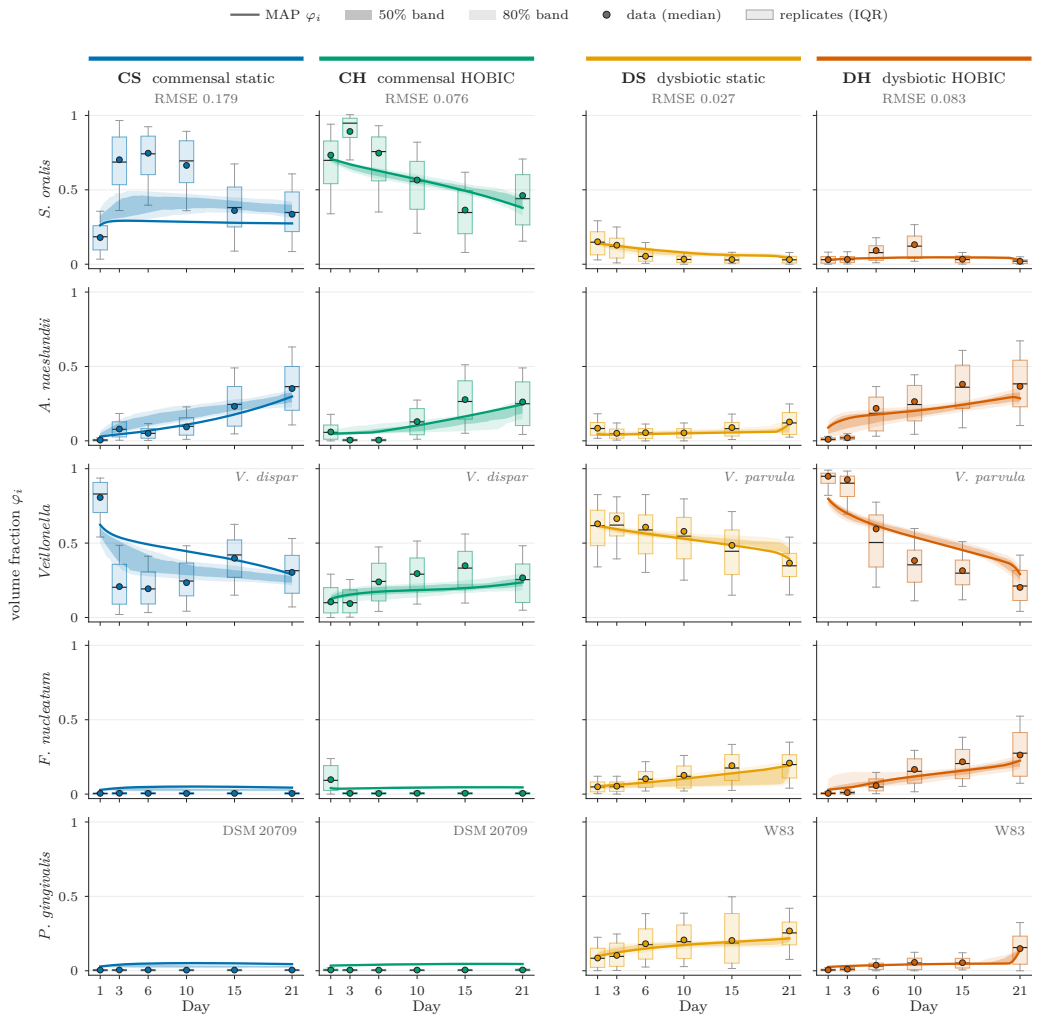


Figure 2: Species fractions over 21 days in the four conditions (columns) for the five taxa (rows). Lines: MAP of stage (iv), species volume fraction ϕ_i; bands: central 50% and 80% posterior predictive intervals pooled over three seeds; dots and boxes: median and interquartile range read off the box plots of Heine et al. [2]. The RMSE of the MAP is given above each column. The *Veillonella* species and the Pg strain differ between the commensal (CS, CH) and the dysbiotic (DS, DH) community (labels in the panels). **[DS and DH from stage (iii) until stage (iv) is accepted.]**

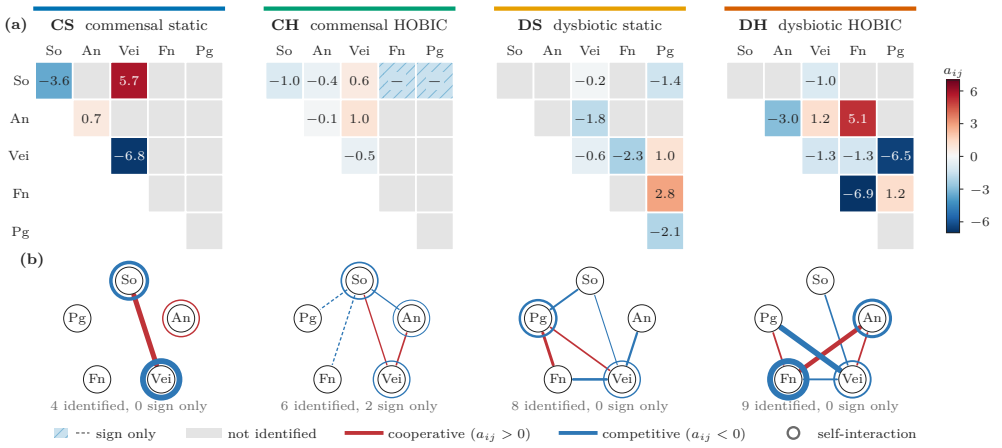


Figure 3: What the data determine, at the final stage with three seeds pooled. (a) Upper triangle of **A** in the four conditions. Coloured with value: identified, i.e. constrained by the data beyond the prior box (posterior SD below 0.8 times that of a uniform distribution on the prior box of the final stage, Supplementary Table S4); colour and number give the posterior median; hatched: only the sign is determined (90% interval excludes zero); grey: not identified. (b) The same entries as a network: red, $a_{ij} > 0$; blue, $a_{ij} < 0$; line width proportional to |median a_{ij}|; dashed, sign only; rings, identified self-interactions a_{ii}. Entries that are not identified are not drawn. **[DS, DH from stage (iii).]**

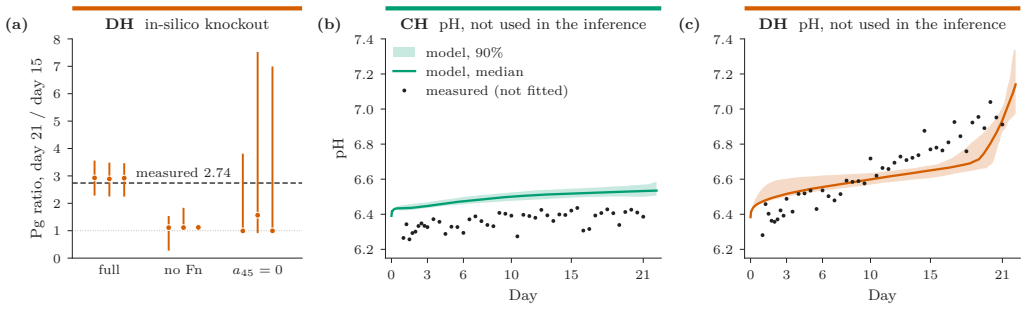


Figure 4: Checks against data and predictions not used in the inference. (a) DH: ratio of the Pg fraction on day 21 to day 15 for the calibrated model, with Fn removed, and with $a_{45} = 0$ (posterior median and 90% interval, one bar per seed; dashed: measured). (b,c) pH predicted from the posterior composition through a regression fitted once on the measured pH and composition (band: central 90% interval), against the measured pH. **[DH from stage (iii).]**